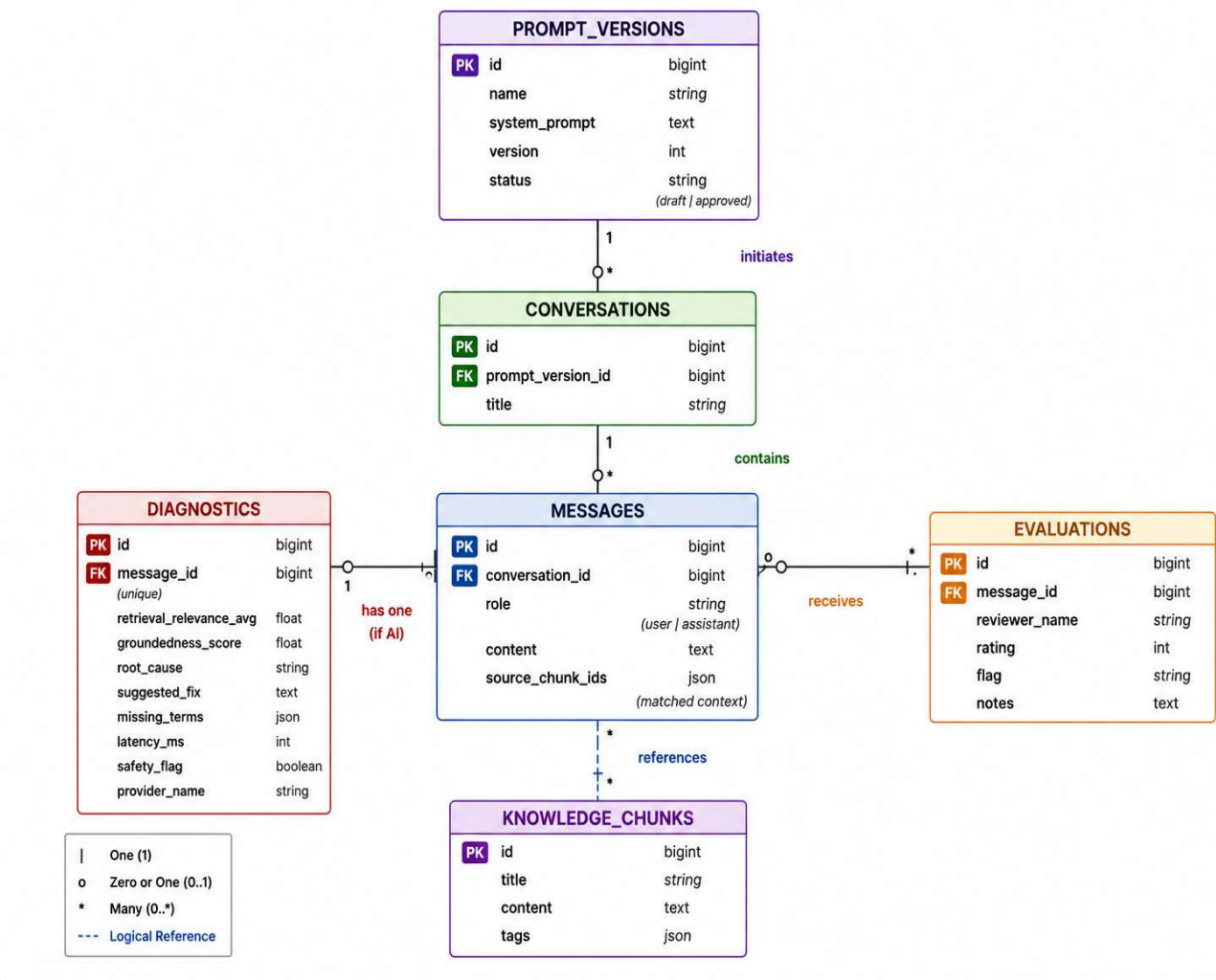


# Lumen AI - Database Design Documentation

This document outlines the relational database schema supporting the Lumen AI Observability platform. The architecture tracks conversations, RAG (Retrieval-Augmented Generation) context, automated diagnostic evaluations, and manual human reviews.

## Entity Relationship Diagram (ERD)

The core structure links `Conversations` and `Messages` to the actual `Prompt Versions` used during execution. `Diagnostics` and `Evaluations` are mapped 1:1 and 1:N to AI messages to track system health and human feedback, respectively. `Knowledge Chunks` operate independently as the vector ground truth.



# Schema Dictionary

## 1. knowledge\_chunks

Stores the ground-truth factual documents used by the Retrieval-Augmented Generation (RAG) system to answer user queries.

| Column     | Type      | Constraints                 | Description                                                     |
|------------|-----------|-----------------------------|-----------------------------------------------------------------|
| id         | BigInt    | Primary Key, Auto-increment | Unique identifier.                                              |
| title      | String    | Required                    | The title or topic of the factual chunk.                        |
| content    | Text      | Required                    | The raw informational content/document text.                    |
| tags       | JSONB     | Nullable                    | Array of tags for domain-filtering (e.g. ["policy", "refund"]). |
| created_at | Timestamp |                             | Record creation time.                                           |
| updated_at | Timestamp |                             | Record update time.                                             |

## 2. prompt\_versions

Manages system instruction versioning. This enables the "Comparative Replay" feature by storing previous and current LLM behavior rules.

| Column        | Type    | Constraints                 | Description                                               |
|---------------|---------|-----------------------------|-----------------------------------------------------------|
| id            | BigInt  | Primary Key, Auto-increment | Unique identifier.                                        |
| name          | String  | Required                    | Descriptive name of the prompt (e.g. "Support Agent v1"). |
| system_prompt | Text    | Required                    | The actual raw system instructions passed to the LLM.     |
| version       | Integer | Required                    | Numeric version tracker.                                  |
| status        | String  | Default: draft              | Enum: draft or approved.                                  |

## 3. conversations

The top-level container grouping a series of back-and-forth messages between a user and the AI.

| Column            | Type   | Constraints                   | Description                                              |
|-------------------|--------|-------------------------------|----------------------------------------------------------|
| id                | BigInt | Primary Key, Auto-increment   | Unique identifier.                                       |
| prompt_version_id | BigInt | Foreign Key (prompt_versions) | The system prompt active when this conversation started. |
| title             | String | Nullable                      | Optional auto-generated chat title.                      |

## 4. messages

Individual dialogue turns (user prompts and assistant completions) within a conversation.

| Column           | Type   | Constraints                 | Description                                            |
|------------------|--------|-----------------------------|--------------------------------------------------------|
| id               | BigInt | Primary Key, Auto-increment | Unique identifier.                                     |
| conversation_id  | BigInt | Foreign Key (conversations) | Links to the parent conversation. Cascades on delete.  |
| role             | String | Required                    | Enum: user or assistant.                               |
| content          | Text   | Required                    | The raw text payload of the message.                   |
| source_chunk_ids | JSONB  | Nullable                    | Array of knowledge_chunks IDs retrieved for this turn. |

## 5. diagnostics

Stores automated observability and guardrail metrics generated *per AI response*.

| Column                  | Type   | Constraints                    | Description                                                 |
|-------------------------|--------|--------------------------------|-------------------------------------------------------------|
| id                      | BigInt | Primary Key, Auto-increment    | Unique identifier.                                          |
| message_id              | BigInt | Foreign Key (messages), Unique | 1:1 relationship to the assistant's message.                |
| retrieval_relevance_avg | Float  | Required                       | Semantic relevance score of fetched context.                |
| groundedness_score      | Float  | Required                       | Score (0.0 to 1.0) indicating factual overlap with context. |
| root_cause              | String | Required, Indexed              | Enum: healthy, knowledge_gap, hallucination, etc.           |

| Column        | Type    | Constraints    | Description                                                   |
|---------------|---------|----------------|---------------------------------------------------------------|
| suggested_fix | Text    | Nullable       | Automated suggestion for remedying the anomaly.               |
| missing_terms | JSONB   | Nullable       | Keywords from user query missing in context.                  |
| latency_ms    | Integer | Required       | Total response generation time in milliseconds.               |
| safety_flag   | Boolean | Default: false | True if the response violated safety or toxicity rules.       |
| provider_name | String  | Required       | Which LLM service served the request (e.g. mock, openrouter). |

## 6. evaluations

Stores manual ratings and notes left by human auditors (used for fine-tuning datasets and RLHF).

| Column        | Type    | Constraints                 | Description                                            |
|---------------|---------|-----------------------------|--------------------------------------------------------|
| id            | BigInt  | Primary Key, Auto-increment | Unique identifier.                                     |
| message_id    | BigInt  | Foreign Key (messages)      | 1:N relationship. A message can have multiple reviews. |
| reviewer_name | String  | Nullable                    | Name of the human auditor.                             |
| rating        | Integer | Nullable                    | Score from 1 to 5.                                     |
| flag          | String  | Nullable                    | Categorical feedback (e.g. good, hallucination).       |
| notes         | Text    | Nullable                    | Free-form review comments.                             |

## Key Indexing Strategy

To ensure query performance for the Traces Dashboard and Health Analytics:

- `diagnostics.root_cause`: Indexed to quickly count health anomalies (e.g., aggregating total `knowledge_gap` cases).
- `diagnostics.message_id`: Unique index enforcing a 1-to-1 strict relationship.
- `messages.conversation_id`: Indexed to rapidly retrieve timeline histories.